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Motion Control of Single Link Flexible Joint Robot Manipulator Using ANFIS MATLAB Simulation

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ABSTRACT

The motion control of single link flexible joint robot manipulator is reduce the vibration of tip of the manipulator. The advance methodology is to optimize fuzzy logic controller parameters via neural network and use the adaptive neuro-fuzzy scheme to control single link flexible joint robot manipulators. The dynamics of single link robot manipulators are highly nonlinear with strong couplings existing between joints and are frequently subjected to structured and unstructured uncertainties. The increased complexity of the motions of robots manipulator considering joint elasticity makes conventional control strategies complex and difficult to synthesize. This paper presents investigations into the development of adaptive neuro Fuzzy control for position and velocity control of a flexible joint manipulator. To study the effectiveness of the controllers, a adaptive Neuro Fuzzy Controller is developed for tip angular position and velocity control of a single link flexible joint manipulator. This is then extended to incorporate an ANFIS Controller for velocity error reduction of the flexible joint system. Simulation results of the response of the flexible joint manipulator with the ANFIS controllers are presented in time domains. The performances of the ANFIS schemes are examined in terms of input position and velocity tracking capability, level of robot vibration reduction and time response specifications.

Keywords: Flexible structure, Manipulator, ANFIS, Uncertain system.

I. INTRODUCTION

In most existing robotic manipulators, maximizing stiffness to minimize vibration and achieve good position and velocity accuracy of robotic manipulators is a key element in their design. This high stiffness is achieved by using heavy material and a huge design. Hence, the existing heavy rigid manipulators are shown to be not much efficient, consuming more power and operational speed. In industry improve the productivity, so that reducing the weight of the arms and increasing their speed of operation are required. Therefore, single link flexible-joint manipulators have received a thorough notice lately, thanks to their lightweight, lower cost, larger work capacity, better manoeuvrability, high speed of operation, more power efficiency, and more number of applications.

However, controlling flexible joint manipulator systems still faces numerous challenges that need to be addressed before they can be used in profusion in everyday real-life applications. The control issue of the single link flexible joint is to design the controller so that link of robot can reach a desired position and velocity for their tracking a given trajectory precisely with low vibration to the link. In order to achieve these objectives, various methods using different kind of technique have been proposed. However, controlling flexible joint manipulator systems still faces many challenges that need to be addressed before they can be used in profusion in everyday real-life applications. The severe type of nonlinearities, coupling stemming from the manipulator's flexibility link, varying different environment operating conditions, structured and unstructured dynamical uncertainties, and external disturbances, are among the typical challenges to be faced with when dealing with such often ill- defined systems.

These kind of Industrial single link flexible robot manipulators are mainly positioning and handling devices. The essential problem in controlling robots is to make the robot manipulator follow a desired input trajectory. In general degree of freedom rigid robot manipulator is characterized by nonlinear, dynamic, coupled differential equations. The problem of controlling robot manipulators still offers much kind of practical and theoretical

challenges due to the complexities of the robot dynamics and the requirement to attain high precision trajectory tracking in the cases of high velocity movement and highly varying loads.

This paper presents investigation into the development of ANFIS for trajectory tracking of tip angular position and velocity control of single link flexible joint manipulator. Initially an ANFIS is developed for trajectory tracking of tip angular position and velocity. The performances of the composite control schemes are examined in terms of input tracking capability, reduction of vibration and time response specifications. The rest of the paper is structured as follows: Section II provides a brief description of the single link flexible joint manipulator system considered in this study. In Section III, we introduce a number of soft computing-based controllers. The design of the proposed controller is detailed in Section IV. In Section V, simulation results are reported and discussed. We conclude with a few remarks and suggestions for further studies pertaining to this important, yet complex, control problem.

II. FLEXIBLE-JOINT MANIPULATOR DYNAMICS

A. Single-link flexible manipulator

Single link flexible manipulator is made of two aluminium strips coupled by ten regularly spaced rigid frames. A DC motor is used an as actuator and 1250 pulses/rev encoder with a one-turn potentiometer measures the angular rotation. The SLFM can rotate 360◦. The sensor to measure the end-point deflection is rather unique. It is a combination of an LED at the end-point and a mirror-detector at the hub. The mirror is centered on the hub axis and is rotated by a dc motor. The light from the LED can fall on the detector only at one position of the mirror and that gives the end-point deflection.

B. Flexible-Joint Manipulator Modeling

Typically, a flexible joint can be modeled as shown in Fig. 1.The actuator is coupled to a flexible transmission through an r : 1 reduction gear. The transmission is directly linked to the load (e.g., manipulator link).Consider a robot manipulator with n revolute flexible joints. Using Euler– Lagrange formulation and neglecting gyroscopic effects, the dynamic equations of the manipulator can be written as:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau_t - \tau_{fl} - \tau_{dl} \quad (1)$$

$$j_m\ddot{\theta} = \tau_m - \frac{1}{r}\tau_t - \tau_{fm} - \tau_{dm} \quad (2)$$

$$\tau_t = K\left(\frac{\theta}{r} - q\right) \quad (3)$$

Where,

$q = R^n$ vector of links' positions; $\theta = R^n$ vector of motors' positions; $M(q) = R^{n\times n}$ manipulator's positive definite inertial matrix; $C(q, \dot{q}) = R^{n\times n}$ matrix of Coriolis and centrifugal terms; $G(q) = R^n$ vector of gravitational torques; $J_m = R^{n\times n}$ motors' diagonal inertial matrix; $\tau_t = R^n$ vector of

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transmission torques; τ_m = R^n motors' generalized torque vector (control input); τ_{fl} =R^n load friction vector; $\tau_{fm} = R^n$ motors' friction vector; dl = R^n load's unmodeled dynamics and external disturbance vector; τ_{dm} = R^n Motors' unmodeled dynamics and external disturbance vector; $K = R^n$ Diagonal matrix of joints' stiffness coefficients; r =R Gear ratio.

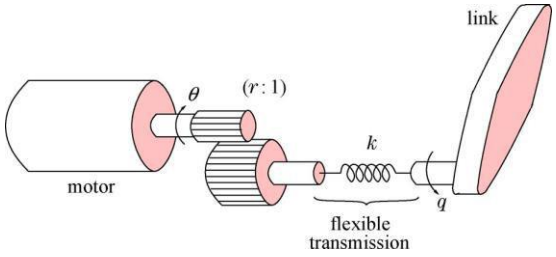


Fig.1. Flexible-joint model

Given the desired trajectories q_d and $\dot{q}_d$, we aim to design a control law τ_m which ensures that the manipulator's position q and velocity $\dot{q}$ track their desired trajectories under unknown dynamics and in the presence of external disturbances. The proposed controller uses q, $\dot{q}$, and $\dot{\theta}$ as system's measurable states, and the manipulator's parameters $M(q)$, $C(q, \dot{q})$, $G(q)$, J_m, τ_{fl}, τ_{fm}, τ_{dl}, and τ_{dm} are assumed to be unknown.

C. Problem Statement

The control objective is to design a control law τ_m to force the manipulator's position q and velocity $\dot{q}_d$ to track their give input time-dependent desired values q_d and $\dot{q}_d$, respectively. This objective is to be achieved under the systems unknown or uncertain of dynamics.

III. SOFT-COMPUTING-BASED CONTROL

In the recent techniques of nonlinear control systems in that area, the common point still shared by the enormous majority of conventional control techniques is their dependence on exact mathematical models of the systems to be controlled for them to provide suitable action. In real life, and due to the typical high nonlinearities within the dynamics of single link flexible-joint manipulators, deriving a precise model for such systems could be a not easy to undertaking. Although conventional adaptive control strategies, such as in sliding mode controllers, compensate for the system's parametric uncertainties, they are still susceptible in the face of unstructured modeling uncertainties. Expert controllers based on tools of soft computing, on the other hand, may not have such a drawback. In fact, computational intelligence tools, in general, have been accredited in a numeral of applications to provide suitable results in the face of relatively large magnitudes of noise in the input signals, of dynamically changeable parameters, and in the require of a precise mathematical model of the system in hand. There are two types of computational intelligence tools that which are we concerned with in this work are fuzzy logic controllers (FLCs) and artificial neural networks (ANNs).

Among the main features of FLCs is their capability to generate sufficient control decisions inference through human- like linguistic descriptions. These are represented by fuzzy rules based on heuristics, knowledge, and

experience, and are often used to control a given system. A particular inference mechanism processes the information stored in the knowledge base to determine the sufficient control action to be taken at the given operating condition.

ANNs represent a one of the type of soft computing technique and an important class of mathematical learning tools known as connectionist modeling. An ANN is a set of interconnected computational nodes in which information is processed and transferred from one node to another through the means of weighted links with the reason of mimicking the functionality of the neurons in the human brain. ANNs are characterized by their nonlinear behavior, parallel processing, and their regular optimization and learning capabilities. These advantages have been behind the increasing popularity of ANNs for numerical modeling and control, especially for systems on which little is known with reference to their dynamics and operating environments. In the same way of FLCs, the neural network universal approximation theorem guarantees that any adequately smooth function can be approximated to any degree of accuracy using a single- hidden-layer ANN. Although several neural network-based controllers have been proposed in the literature, the supervised multi-layer perceptron scheme is among the simplest and most popular schemes, particularly in control systems' applications. The network's learning mechanism is often accepted out as to minimize the network's output error based on a user-defined feedback signal

IV. ADAPTIVE NEURO FUZZY CONTROL

In recent days, intelligent control in general, and adaptive neuro-fuzzy control in particular, have been quite inspiring paradigms for real-time various control applications. Neuro- control based artificial neural networks, having the capacity to learn from input-output non-linear functions, are best candidates for solving more complex nonlinear control problems. Neurons are basically non-linear elements; hence that neural networks are basically non-linear systems which can be used to learn and solve non-linear control problems that are usually so difficult for traditional and conventional control methods to handle. Using the inverse model as the main block in the neuro-control approach is one of the most widely applied schemes. In order to achieve accurate trajectory tracking and best control performance, a number of control schemes have been developed. ANFIS is a fuzzy inference system implemented in the framework of adaptive networks. Hybrid learning procedure use of the ANFIS can construct an input- output mapping based on both human knowledge (in the form of fuzzy if-then rules) and stipulated input-output data pairs. The hybrid learning algorithm identifies the membership function parameters of single-output, Sugeno type fuzzy inference systems (FIS). A combination of least mean squares (LMS) and back propagation gradient descent methods are used for training FIS membership function parameters to model a given set of input/output data. The parameters associated with the membership functions change through the learning process. The training process stops whenever the designated epoch number is reached or the training error goal is achieved.

The nonlinearity prevailing in the arm dynamics induces high uncertainty in the performance of the robotic manipulators under conventional control strategies. The use of the intelligent systems such as neural networks and fuzzy control has provided better results. But a combination of such intelligent systems, like, neuro-fuzzy or ANFIS provides even better results than just neural networks.

A. ANFIS architecture

For simplicity, we assume the fuzzy inference system under consideration has two inputs x and y and one output z. Fig. 3 Depicts the structure of Adaptive neuro fuzzy controller below:

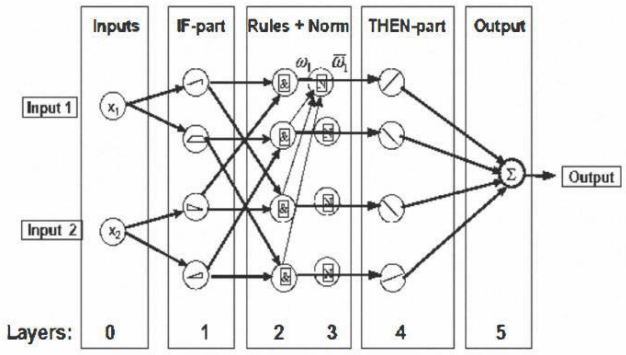


Fig.2. Structure of neuro fuzzy controller

B. ANFIS Modeling

Modeling was performed using MATLAB 7.8. ANFIS and Sugeno-type fuzzy inference systems were used in the modeling of robot manipulator. Single Input Single Output models consisting of inputs including position and velocity were developed to predict outputs. Sugeno-type fuzzy inference systems were generated by using Genfis which utilized subtractive clustering to compute the models for the product properties.

The purpose of clustering was to identify natural groupings of data to produce a concise representation of the behavior of the system. The fuzzy models generated from the membership functions and rules were data-driven by the process data for each mechanical property. Each set of process data collected from the extrusions consisted of 40 data points from which 70% and 30% were selected randomly for training and testing, respectively. The models were developed and implemented using 3000 epochs and a radius of 0.5. The input data were the process data acquired by the computer consisting position and velocity readings from extrusion. The input and output data sets contained inputs (position, velocity) and one output torque.

V. SIMULATION RESULTS AND DISCUSSION

Experimental Setup

Let $\Delta q = q - qd$ and $\Delta\mu = \mu - \mu d$ denote the links' and motors' position errors, respectively, with μd being the unknown desired time-dependent motor position vector. The control strategy is based on the design of an adaptive controller that not only leads to a precise tracking of the system's nominal desired signals, but also improves the motors' internal stability. Should the motors' desired position μd have been available, the control strategy would be based on tracking Δq and $\Delta\mu$ to zero. Since that is not the case, we define the following compounded velocity error signal $\Delta\epsilon_r = \dot{q}_d - (\lambda_q + (1-\lambda)\frac{1}{r}\dot{\theta})$.

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For a diagonal matrix $\lambda = \text{diag}(\lambda 1, \lambda 2 \ldots\ldots \lambda n)$ with $\lambda i \in [0,1], i = 1 \ldots\ldots n$. The feedback gain λ is introduced to provide a trade off between the link tracking performance and internal stability, due to the high nonlinear coupling between the two.

To demonstrate the performance of the proposed controller, a set of numerical experiments is carried out on a single link flexible-joint manipulator. The manipulator's dynamics in terms of its physical parameters is defined by $M(q) = I$, $C(q, \dot{q}) = 0$, and $G(q) = mgl \sin(q)$, where m is the link's mass, g is the gravity constant, and l is the link's length. Table I summarizes the manipulator's physical parameters along with their respective values. The stiffness coefficient and gear ratio are assumed to be $K = 5$ N • m/rad and $r = 1$.

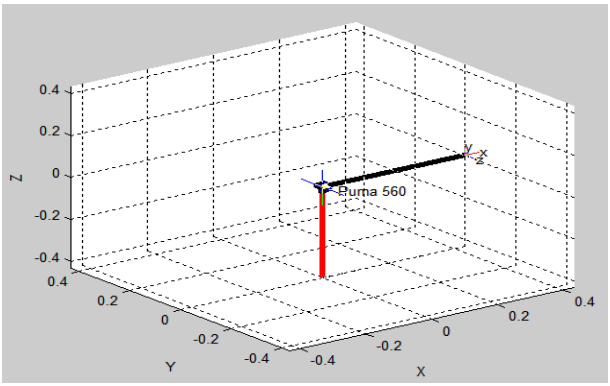


Fig.3 simulation diagram of single link flexible robot manipulator

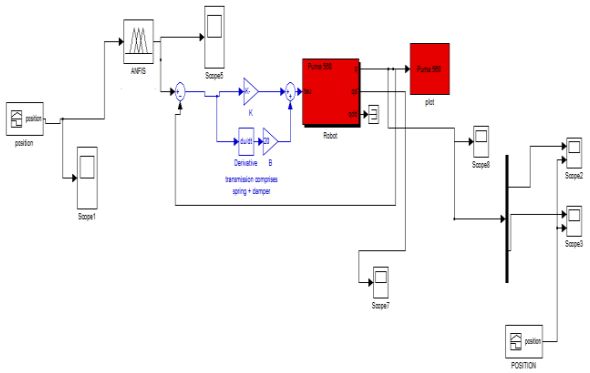


Fig.4 Model of single link robot manipulator position control simulink

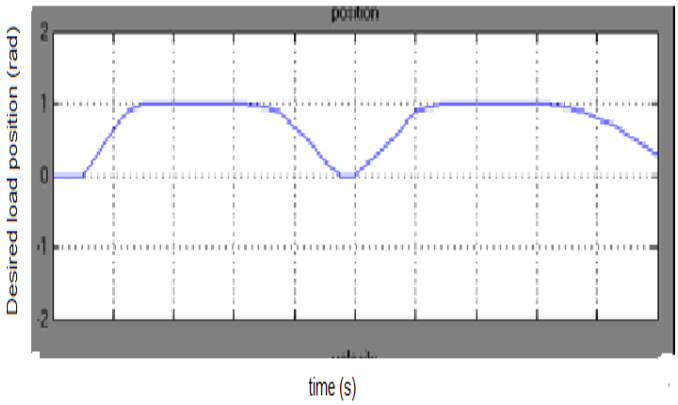


Fig.5 Manipulator's position reference signals.

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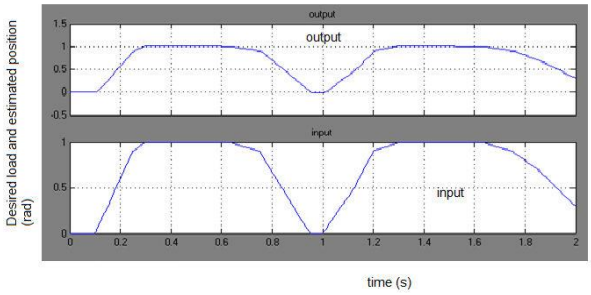


Fig. 6 output of single link flexible robot manipulator position

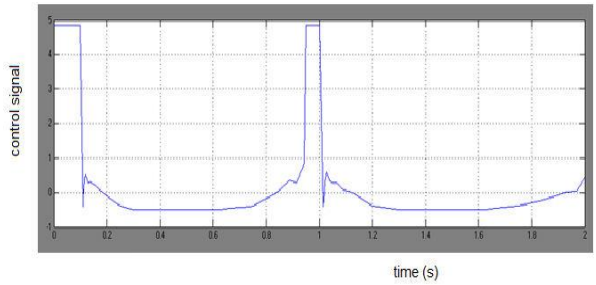


Fig.7 Anfis controller response of manipulator position

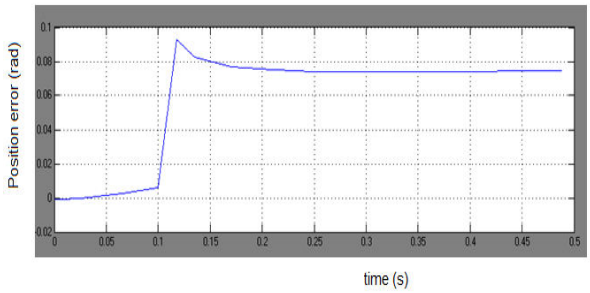


Fig.8 Single link manipulator's Position error

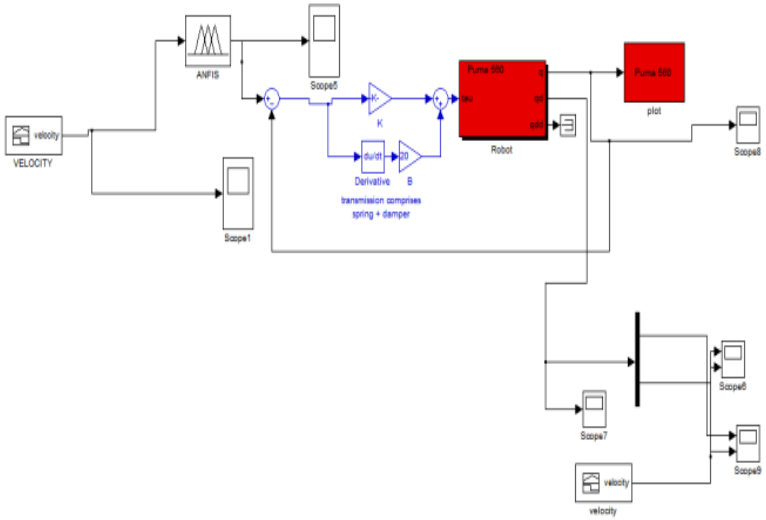


Fig.9 Model of single link robot manipulator velocity control simulink

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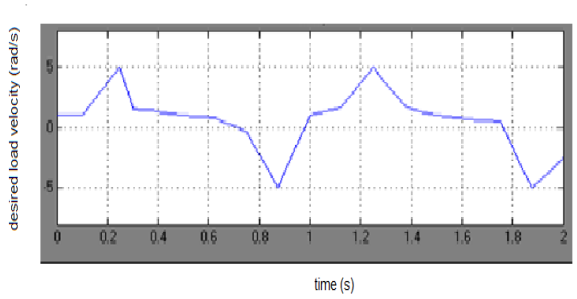


Fig.10 Manipulator's velocity reference signals.

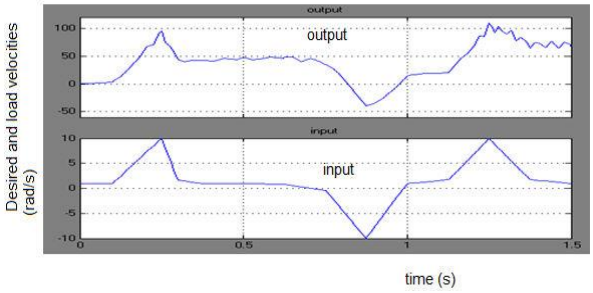


Fig.11 output of single link flexible robot manipulator velocity

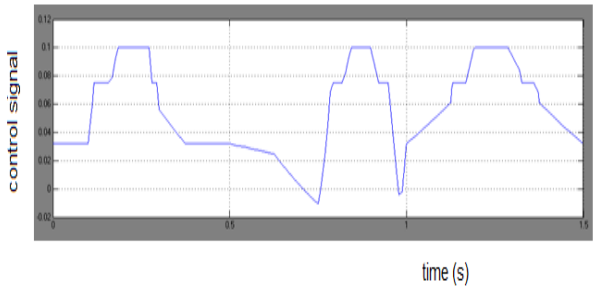


Fig.12 ANFIS controller response of manipulator velocity

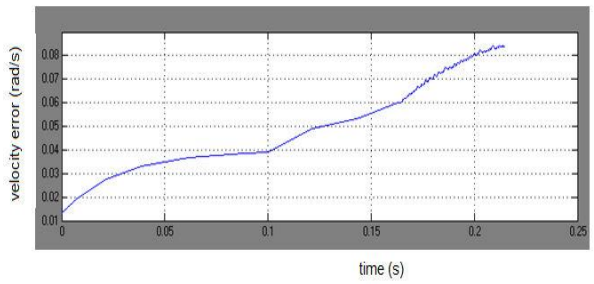


Fig 13 Single link manipulator's Velocity error

VI. CONCLUSION

The development of adaptive neuro fuzzy control techniques for position and velocity error reduction of a flexible joint manipulator has been presented. The proposed control schemes have been implemented through MATLAB Simulink. The control strategy is based on a Adaptive neuro fuzzy control approach while taking into account the actuators relative stability criterion by introducing a trade-off between the actuators' internal stability

and the link's position. The performances of the control schemes have been evaluated in terms of input position and velocity error reduction at the resonance modes of the manipulator. Acceptable performance in input position and velocity control has been achieved with proposed control strategies. The work has developed and reported in this forms the basis of design and development of hybrid control schemes for input position and velocity error reduction of multi-link flexible manipulator systems and can be extended to and adopted in practical applications.

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